

Fine-Grained Spatial–Temporal Gait Recognition Network Based on Millimeter-Wave Radar Point Cloud

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Abstract—Radar-based gait recognition has gained wide attention recently for its ability to preserve privacy and adapt to low-light and poor-weather scenarios. Among different forms of input data, a radar point cloud is an appealing option as it captures not only the appearance signatures but also the motion signatures of the subject. For gait recognition, both appearance and motion are crucial signatures that can be represented by spatial features and temporal features, respectively. However, the spatial–temporal features extracted by existing radar point cloud-based methods are coarse-grained, leading to poor performance in realistic applications. To enhance the spatial–temporal feature representation ability of the radar point cloud-based method, in this article, we design a novel network to extract fine-grained spatial–temporal gait features from a millimeter-wave (mmWave) radar point cloud. For fine-grained spatial feature extraction, we apply a dual-stream feature extraction (DSFE) module to exploit the 3-D coordinates, intensity, and velocity information within the radar point cloud. After that, since each body part has its unique characteristics in the gait task, we also propose a probability-guided body-part partition (PGBP) module to split the radar point cloud into fine-grained spatial body parts. For fine-grained temporal feature extraction, a local–global temporal feature extraction (LGTE) module is used to further capture the temporal patterns of each body part. To evaluate the effectiveness of the proposed methods, we conduct extensive experiments with 90 subjects in various realistic settings, i.e., cross-view and cross-wearing conditions. The results demonstrate that our model achieves significant improvement over existing radar-based gait recognition methods.

Index Terms—Body-part partition, gait recognition, local–global temporal modeling, radar point cloud, spatial–temporal modeling.

I. INTRODUCTION

GAIT recognition is an innovative biometric recognition technology that aims to identify pedestrians by their walking patterns [1]. Compared with other biometric modalities, e.g., face [2], fingerprint [3], and iris [4], gait can be captured from a long distance and requires no cooperation

of subjects, making it hard to disguise. As a result, it has been deployed in numerous scenarios, such as crime investigation, access control, and social security [5]. Currently, various sensors have been considered in gait recognition systems, such as wearable sensors [6], camera [7], LiDAR [8], WiFi [9], and radar [10]. Among them, the radar-based method gains plenty of attention due to the advantages of adaptability to illumination, privacy preservation, and high resolution.

Existing radar-based gait recognition methods mainly depend on micro-Doppler (m-D) signatures to identify people [10], [11], [12]. During walking, besides the bulk motion induced by the torso, some other body parts, e.g., limbs, legs, and arms are also rotating or swinging, which could induce additional frequency modulations on radar echoes, known as the m-D. m-D reflects the unique micro-motion patterns of each person, which proved highly significant for gait recognition. Many articles perform time–frequency analysis to obtain time-Doppler spectrogram for m-D information from radar echo. However, Doppler frequency can be easily affected by the angle between the target moving direction and radar line of sight (LOS) [13]. That means the time-Doppler spectrogram will be undermined seriously due to the change of the walking view, especially when people walk perpendicular to radar LOS. Moreover, the time-Doppler spectrogram lacks a comprehensive description of appearance patterns, resulting in limited gait information.

With the rapid development of automatic drive and 5G technologies, millimeter-wave (mmWave) radar has become an appealing remedy for fine-grained sensing [14]. Based on the superior resolution, mmWave radar can generate a highly informative radar point cloud, which contains the 3-D location, velocity, and intensity of the target’s strong scattering point. As shown in Fig. 1, radar point cloud sequences incorporate both temporal motion signatures and spatial appearance signatures, which turn out to be more abundant in gait information than time-Doppler spectrograms. Besides, appearance is less susceptible to variations in walking view in contrast to Doppler frequency. Motivated by this, in this article, we aim to improve the accuracy of radar-based gait recognition methods based on mmWave radar point cloud.

Blessed with the power of feature learning, a deep neural network is commonly used by existing gait recognition methods to extract discriminative gait features. And there have been some researches on mmWave radar point cloud-based

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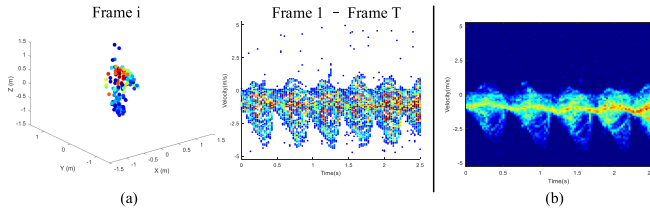


Fig. 1. Illustrations of (a) mmWave radar point cloud and (b) time-Doppler spectrogram. Each radar point contains its 3-D coordinates, intensity, and velocity. Point cloud sequences contain both spatial appearance and temporal motion signatures of the subject, while the time-Doppler spectrogram lacks a comprehensive description of the appearance pattern.

gait recognition [15], [16], [17]. However, the spatial and temporal feature extraction way of previous works is generally coarse-grained. Hence, they have not fully utilized the abundant information available in radar point clouds. From the perspective of spatial modeling, first, their point feature extraction way is coarse-grained. Concretely, most of them simply apply existing deep convolutional neural networks (CNNs) or complex point feature extraction networks to extract point-level features, which may not be suitable for sparse radar point clouds. Besides, measuring Doppler velocity is a special advantage of radar. That is, each point owns its accurate Doppler velocity that represents the motion pattern, which is instructive for the gait recognition task. Existing methods lack the ability to fully leverage 3-D coordinates, intensity, and velocity information and then fuse them effectively. To summarize, it is still a challenge to extract representative point-level features from radar point clouds. Second, existing methods typically treat points within a single frame as a whole, and aggregate all pointwise features into a vector. As we know, there are significant differences between human body parts in terms of motion and appearance patterns during walking. Treating the human body as a whole is a coarse-grained strategy and will induce information loss. We are supposed to handle each body part differently. Splitting the feature map into strips has been commonly used in the person Re-ID field [18]. However, the radar point cloud is sparse in contrast to the visual point cloud. The point number in a single frame is small, which means radar point clouds may only contain a subset of body parts. Body parts with small radar cross sections (RCSs), e.g., legs and arms, could only reflect low-energy radar signals, resulting in little point clouds. The mmWave radar could only capture part of appearance information within a single frame, less comprehensive than a camera and LiDAR system where every snapshot shows all body parts. Thus, we cannot apply existing partition methods to radar point clouds directly. How to divide radar point clouds into fine-grained body parts reasonably remains a challenging problem. From the perspective of temporal modeling, current methods have yet to take full advantage of the temporal motion information. These methods usually use long-short-term memory (LSTM) [19] or transformer [20] network to model temporal changes of motion. They focus on long-range dependencies of gait but ignore the fine-grained motion within short time scales. In conclusion, available models perform poorly in extracting fine-grained and discriminative features

from sparse radar point clouds, leading to poor performance in realistic applications.

Furthermore, gait recognition is commonly perceived as a retrieval task in real-life applications [5]. Compared with the classification-based approach, the extendibility of the retrieval-based approach is better. Generally, retrieval-based approaches rely on similarity measurement rather than classification probability to realize recognition. Retrieval-based approaches need to build a gallery that contains the samples of individuals to be recognized. Given a test sample (probe sample), we first calculate the feature distance between the probe sample and all gallery samples and then assign the identity of the most similar gallery sample to the test sample. As a result, the identities of the training and testing set can be different, and our model is able to recognize a new individual by simply adding a sample of this individual into a gallery. Unfortunately, there are few radar-based methods focusing on retrieval tasks except for [21], [22]. Reference [22] is a spectrogram-based method thus making it susceptible to the variations in walking view. Even though [21] has made some progress based on the radar point cloud, its feature extraction and temporal modeling are still defective and coarse.

Motivated by the above findings, in this article, we propose a novel network to extract fine-grained spatial-temporal gait features from mmWave radar point cloud based on a retrieval task. We apply a mmWave radar to record an echo signal followed by a signal process pipeline to generate a radar point cloud. For fine-grained spatial feature extraction, we first propose a dual-stream feature extraction (DSFE) module with a multilayer perceptron (MLP). The dual-stream network finely exploits RCS and velocity information separately based on a spatial-RCS pathway and a spatial-velocity pathway. Besides, our DSFE applies a multiscale feature fusion strategy to effectively fuse the output features of two pathways. Concerning that different human body parts possess diverse appearance and motion patterns, we also introduce a probability-guided body-part partition (PGBP) module to split sparse radar point clouds into fine-grained spatial body parts. In brief, we divide each body part based on the spatial probability distribution of the human radar point cloud. For fine-grained temporal feature extraction, we design a local-global temporal feature extraction (LGTE) module to further capture the temporal motion pattern of every body part. Our temporal module can extract local temporal features with different granularities and aggregate local features to produce global temporal features based on attention summation. Based on three key modules, our network is reliable in extracting discriminative gait features from the radar point cloud. To evaluate the performance of the proposed method in realistic scenarios, we have collected radar data from 90 subjects under ten views and three wearing conditions and conducted extensive experiments in cross-view and cross-wearing-condition settings. The results show that our method performs significantly better than existing radar-based gait recognition methods.

The main contributions of this article effort are as follows.

- 1) To extract fine-grained spatial features from the radar point cloud, we propose a DSFE module and a PGBP module. DSFE leverages two pathways to finely exploit

the RCS and velocity information within each radar point. PGBP is designed to obtain fine-grained representations of different body parts for sparse radar point clouds, which is based on the spatial probability distribution of radar point clouds.

- 2) To further extract fine-grained temporal features for each spatial body part, we propose an LGTE module. LGTE applies temporal convolution with multiple scales to capture local gait features across various temporal granularities. In addition, it incorporates an attention summation module to adaptively aggregate the local features and produce the global temporal feature for each body part.
- 3) We collect gait data from 90 subjects using mmWave radar, including ten walking views and three wearing conditions. In addition, we have conducted extensive experiments in realistic settings, specifically cross-view and cross-wearing-condition scenarios. The comparison results clearly indicate the superior performance of our model over existing radar-based gait recognition methods.

The rest of this article is organized as follows. Section II briefly introduces the related work on radar-based gait recognition and point cloud networks. Section III studies the signal processing pipeline for human point cloud generation. Section IV elaborates on the details of the proposed gait recognition framework. Section V presents the experiments with analysis, and Section VI concludes this article.

II. RELATED WORK

Radar-based gait recognition has been widely investigated for many years, with most studies generally relying on a time-Doppler spectrogram or radar point cloud as input. In this section, we will first discuss these two forms of methods, respectively.

A. Time-Doppler Spectrogram-Based Gait Recognition Methods

Time-Doppler spectrogram could capture the micro motion patterns of different body parts, making it a suitable data format for identifying human gait patterns. There are many works that focus on distinguishing between various types of gait based on m-D signatures [23], [24], [25]. Different from these gait classification methods, our gait recognition task aims to distinguish between different identities without focusing on a specific gait type of target. In the early days, researchers mainly focus on extracting traditional features, e.g., Doppler frequency of torso, Doppler bandwidth, and entropy from spectrogram [26], [27]. However, these hand-crafted features are hard to design and insufficient for complex conditions. Recently, due to the superior ability in automatic feature extraction and feature representation, deep CNN has been widely adopted in various tasks. Cao et al. [28] first applied CNN to extract features from the time-Doppler spectrogram and achieve 85.6% accuracy with ten subjects. Vandersmissen et al. [10] used a low-power FMCW with deep CNN to build a gait recognition system and achieve accuracy

close to 80% over five subjects. Yang et al. [29] designed a multilayer fusion network to combine features at different scales. Li et al. [30] introduced a recognition model based on distributed radar sensor fusion, which can realize sequential gait recognition and falling. Shi et al. [31] proposed to fuse spectrogram and gait energy images to extract more robust gait features, which achieve better performance. Lan et al. [22] published a spectrogram dataset under multiview and multi-wearing conditions and proposed a method based on a retrieval task. Although [22] has considered the change of walking view and wearing condition, their experiment result is suboptimal. To sum up, current spectrogram-based methods have made some progress, but the time-Doppler spectrogram is short in appearance description and vulnerable to the variations of walking view, making the performance of the above methods still limited.

B. Radar Point Cloud-Based Gait Recognition Methods

With the increase in resolution, modern commercial mmWave radar can generate high-resolution human point clouds. And researchers have explored the utilization of radar point clouds for gait recognition in their studies. mID [15] first proposed to use radar point cloud and achieve 89% identification accuracy with 12 subjects. mID leverages a voxelization strategy to transform the point cloud into grid data format, which will induce resolution loss. Later, Meng et al. [16] proposed a CNN-based framework mmGaitNet. They focus on gait recognition under coexisting multiple-people scenes and achieve 88% accuracy for five coexisting persons. Cheng and Liu [21] used PointNet [32] and LSTM to build SRPNet for retrieval-based gait recognition, which achieve 90.1% accuracy for 40 subjects. Li et al. [17] proposed MTPGait for multiperson gait recognition based on CNN and LSTM. STPointGCN [18] builds an end-to-end model for co-existing multiple people recognition based on graph convolutional networks (GCNs). Recently, Huang et al. [33] designed a hierarchical network HDNet to extract spatial-temporal features from radar point clouds and achieve promising results for classification tasks. However, gait features extracted by the existing framework are generally coarse-grained and consequently, these methods have not fully explored the potential of radar point cloud. Besides, they have not investigated the model performance under realistic scenarios, i.e., cross-view and cross-wearing conditions.

III. GENERATION OF HUMAN POINT CLOUD FROM MMWAVE RADAR DATA

A. mmWave Radar Signal Model

Linear frequency modulated continuous wave (LFMCW) is the most common waveform emitted by existing commercial mmWave radar. As shown in Fig. 2, a sequence of chirp signals is transmitted during one frame period.

A chirp is a sinusoid whose signal property is characterized by the start frequency f_c , bandwidth B , and duration time T_c

$$s(t) = \exp \left\{ j2\pi \left(f_c t + \frac{B}{2T_c} t^2 \right) \right\}, \quad 0 \leq t \leq T_c. \quad (1)$$

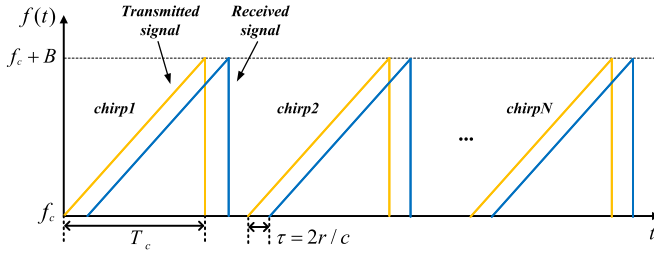


Fig. 2. Chirp sequences within a single frame.

Specifically, the frequency of the chirp signal increases linearly with time: $f = f_c + St$, and $S = B/T_c$ denotes the rate at which the chirp ramps up. Assuming that the distance of the target is r and the velocity of the target is v , the received signal is delayed by time $\tau = 2(r + vt)/c$, where c denotes the speed of light. Generally, we mix the reflected signal with the transmitted signal followed by a low pass filter to get the intermediate frequency (IF) signal. By sampling each IF signal with f_s frequency, we can obtain a 2-D radar data matrix [34]

$$d(n, p) \approx \exp \left\{ j2\pi \left[\left(\frac{2Sr}{c} + f_d \right) \frac{n}{f_s} + f_d p T_c + \frac{2f_c r}{c} \right] \right\} + \omega(n, p) \quad (2)$$

where $f_d = 2f_c v/c$ denotes the Doppler shift frequency and ω denotes the additive white Gaussian noise. n represents the index of sampled data within a single chirp, and p represents the index of chirp. We can apply fast Fourier transform (FFT) along fast time n to estimate the distance of the target within one chirp. After that, another FFT can be applied across slow time p to estimate the velocity of the target within one frame. In brief, we can obtain a range-Doppler (RD) map of the target by 2-D FFT.

However, spatial information is not included in the RD map. We can utilize the technology of array signal processing to estimate the angular location of the target. Existing commercial mmWave radar usually consists of multiple transmitting and receiving antennas, which form an antenna array. Each antenna element has its own signal channel, that is, we can obtain a 3-D radar data cube $d(n, p, l)$, where l represents the index of the antenna channel. Considering there are M targets and L uniformly placed antennas. For target m , the time delay measured at antenna l can be calculated as

$$\tau_{lm} = \frac{2(r_m + v_m t) + l\delta \sin \theta_m}{c} \quad (3)$$

where r_m , v_m , and θ_m indicate the distance, velocity, and azimuth angle of the target, respectively, and δ represents the distance between antennas. We can obtain the final 3-D data cube $d(n, p, l)$ based on (2) and (3)

$$d(n, p, l) \approx \sum_{m=1}^M \exp \left\{ j2\pi \left[\left(\frac{2Sr_m}{c} + f_{dm} \right) \frac{n}{f_s} + \frac{f_c l \delta \sin \theta_m}{c} + f_{dm} p T_c + \frac{2f_c r_m}{c} \right] \right\} + \omega(n, p, l). \quad (4)$$

The RD-angle (RDA) map can be obtained by performing a third FFT on the RD map along antenna channels; RDA contains information about the distance, velocity, and angle estimation of a target. Nevertheless, performing FFT at each bin within the RD map is time-consuming and the output RDA map requires a large amount of storage space. In most cases, the target of interest occupies only a small range within the RD map. Hence, we can apply a target detection algorithm such as a constant false alarm rate (CFAR) to detect high-energy locations before applying angle FFT. By performing detection and angle estimation, we can acquire a set of points, i.e., radar point cloud, which contains the 3-D location, velocity, and intensity of the target's strong scattering points.

B. Overview of the Signal Processing Pipeline

The whole signal processing pipeline for generating human radar point cloud is shown in Fig. 3. We first perform range FFT to get the range profile, and the size of range FFT is set to 128. For static clutter removal, based on the assumption that the radar echo from static objects is likely to remain unchanged among different chirps. After range FFT, we apply the average cancellation algorithm to remove static clutter. Specifically, we first calculate the mean value of the range profile along the slow time axis and subtract the mean value from each range profile. Then, Doppler FFT with a size of 128 is performed to get the RD map. After that, CFAR detection is applied to detect the high-energy locations in the RD map, and the raw point cloud is obtained by applying angle FFT with a size of 256 to CFAR detection results. Before each FFT, a Hamming window is applied along the corresponding dimension. For CFAR detection, we successively apply the cell-average smallest of CFAR (CASO-CFAR) along range bins and Doppler bins. The threshold factor of both CFAR detectors is set to 1.2, while the length of the reference window is set to 8 and 4, respectively. However, due to the multipath effect, i.e., high-order reflection between walking people and static objects, there are still numerous clutter points that do not belong to the walking target in the raw radar point cloud. Given that the real target trajectory is stable during walking while the ghost trajectory induced by multipath is usually erratic. We could apply the moving target tracking algorithm to help remove these clutter points. Specifically, we apply the constant velocity model and Kalman filter (KF) to estimate the target state and associate the current observation state with the existing trajectory based on the Hungarian algorithm. By doing this, the real target centroid position at the current frame can be estimated. Concerning the human body shape, we can select a candidate cube area centered at the target centroid and discard all other points. In order to make the cube area cover all target points, in this article, we set the height of the cube to be 3 m, and the width and length to be 1.5 m. Finally, density-based spatial clustering for application with noise (DBSCAN) algorithm is executed to further extract the clean target point cloud. For the DBSCAN algorithm, the minimum points and the radius are set to 5 and 0.1, respectively. Simply applying DBSCAN is not enough to remove clutter points thoroughly. In contrast, we can tackle

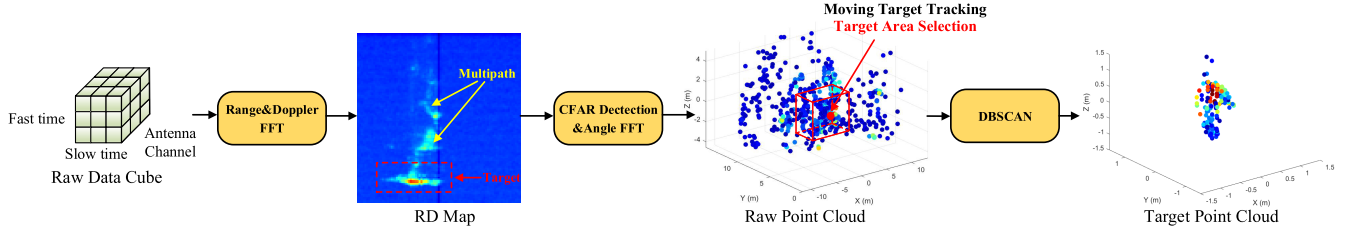


Fig. 3. Signal processing pipeline for generating human radar point cloud.

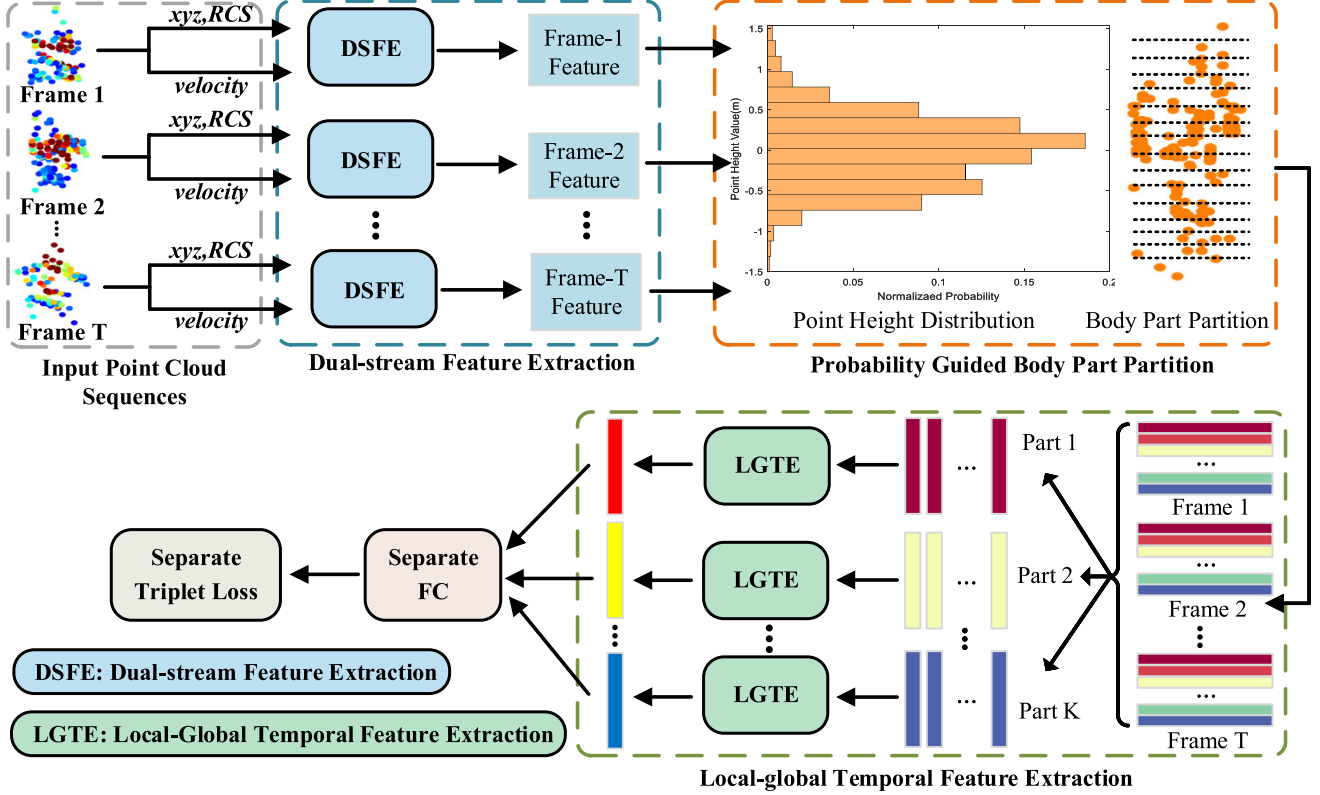


Fig. 4. Framework of proposed method. For each input frame, we apply a DSFE module to fully extract the abundant information within the radar point cloud. After that, frame-level features are divided into different body parts based on the spatial distribution probability of the radar point. And LGTE module is applied to extract the final spatial-temporal part-level temporal features. Separate triplet loss is employed for metric learning.

complex clutter points by combining a tracking algorithm and DBSCAN.

IV. PROPOSED METHOD

A. Network Pipeline

The overview architecture of our method is presented in Fig. 4. The input is the mmWave radar point cloud sequences, which are denoted as $P = \{P_1, P_2, \dots, P_T\}$. T indicates the number of input frames and $P_i = \{p_1^i, p_2^i, \dots, p_N^i\}$ is the set of N points within i th frame. For each point $p_j^i \in \mathbb{R}^5$, it comprises five initial attributes, i.e., 3-D coordinates x_j^i, y_j^i, z_j^i , Doppler velocity v_j^i , and scattering intensity RCS_j^i . And each input frame can be regarded as a two-dimension matrix $P_i \in \mathbb{R}^{N \times 5}$. For later body partition, we sort the points $\{p_j^i | j = 1, \dots, N\}$ within each frame by their height value along the z -axis. That is, the first row of P_i is the highest point in the i th frame while the last row is the lowest point.

Hence, the row dimension of P_i is relevant to the height of the human body.

For each frame P_i , its sorted points are fed into a DSFE module to produce the high-level pointwise feature $F_i \in \mathbb{R}^{N \times C}$, where C indicates the number of feature channels. Afterward, we split the frame-level feature F_i into different body parts based on the spatial distribution probability of mmWave radar point cloud. Specifically, we first split F_i into K regions along row dimension and then aggregate each region's feature to produce spatial part-level feature, which denotes as $F_i^k \in \mathbb{R}^C, k = 1, \dots, K$. Gathering each part feature over T frames, we can obtain the time-varying part feature $B_k = \{F_i^k | i = 1, \dots, T\} \in \mathbb{R}^{T \times C}$. Next, the LGTE module is applied to model the temporal features of each part B_k . We can get the final spatial-temporal part-level features $B'_k \in \mathbb{R}^{C'}$ and $B' = \{B'_k | k = 1, \dots, K\} \in \mathbb{R}^{K \times C'}$, where C' indicates the number of output temporal feature channels. In the end, a separate fully connected (FC) layer

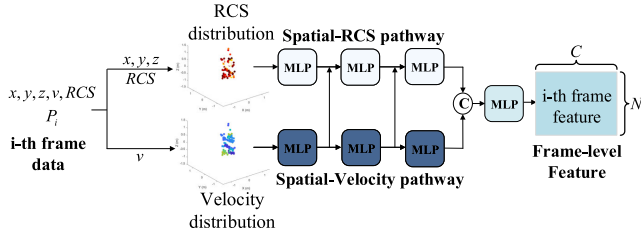


Fig. 5. Detailed structure of DSFE module, which consists of a spatial-RCS pathway and a spatial-velocity pathway. N indicates the number of points and C indicates the number of feature channels.

is employed to map the part-level feature to metric space for gait recognition.

Unlike a typical classification task, our model is based on a retrieval task. There is no overlap between the identities in our training and testing sets. In other words, our trained model is expected to retrieve the gait pattern of a new individual, even if it has not been trained on that individual's gait data. Concretely, during the training stage, we optimize the model to learn the ability to minimize the intrainidentity discrepancy and maximize the interidentity discrepancy with triplet loss [35]. In the testing stage, we measure the similarity between the output feature of the test sample and gallery samples. And assign the identity of the most similar gallery sample to the test sample.

B. Dual-Stream Feature Extraction

As mentioned in Section II, most of the existing point feature extraction methods are complex and mainly designed for dense visual point clouds, which are not suitable for sparse radar point clouds. As a pioneer work, PointNet [32] first proposed to employ shared MLP independently on each point, which is simple but effective. To fully exploit 3-D coordinates, velocity, and RCS information within the radar point cloud, a dual-stream architecture with multiscale fusion is designed based on MLPs. As shown in Fig. 5, our model involves a spatial-RCS pathway to capture the RCS feature and a spatial-velocity pathway to capture the velocity feature, and both of them are composed of MLPs. For input radar point cloud $P_i \in \mathbb{R}^{N \times 5}$, its 3-D coordinates and RCS reflect the appearance pattern of the subject while the Doppler velocity reflects the motion pattern. To fully exploit the abundant information available in the radar point cloud, we handle these two forms of information separately to obtain more fine-grained point feature representation. First, we separate the input data into two parts: $P_i^{\text{spatial}} \in \mathbb{R}^{N \times 4}$ and $P_i^{\text{velocity}} \in \mathbb{R}^{N \times 1}$, where P_i^{spatial} contains the 3-D coordinates and RCS, P_i^{velocity} contains the Doppler velocity. After that, a dual-stream network is applied to extract their high-level feature. Each pathway is composed of several MLPs. Specifically, MLP is a small network composed of an FC layer, a batch normalization layer, and an activation function

$$F_i^l = \text{MLP}_l(F_i^{l-1}) = \text{LeakyReLU}(\text{BN}(F_i^{l-1}W_l)) \quad (5)$$

where F_i^{l-1} and F_i^l denotes the output feature of $(l-1)$ th and l th layer, respectively, $W_l \in \mathbb{R}^{C_{l-1} \times C_l}$ refers to the weights of FC layer. Compared with complex feature extraction networks,

which mainly rely on exploring sophisticated local geometric structure information, MLP treats every point independently and equally. As we know, the radar point cloud is sparse and contains only part of the human body, leading to limited local geometric representation. Thus, directly applying complex methods to radar point clouds may not achieve satisfactory results. We have attempted various point cloud networks in the experiment to investigate this issue, and the result will be discussed in Section V.

Besides, we employ a multiscale feature fusion strategy to fuse features of two pathways effectively. For early fusion, we fuse the information of the two pathways similar to [36]. That is, we fuse the features of the spatial-velocity pathway into a spatial-RCS pathway based on channel concatenation, as the Doppler information is a good complement for spatial information. For deep fusion, we cascade the output of feature two pathways along channel dimension and employ another MLP to produce the final frame-level feature $F_i \in \mathbb{R}^{N \times C}$

$$F_i = \text{MLP}_{\text{fuse}}(F_i^{\text{spout}} || F_i^{\text{dpout}}) \quad (6)$$

where F_i^{spout} and F_i^{dpout} indicates the output of two pathways, $[||\cdot]$ represents the channel concatenation operation. Considering the large difference between the two forms of feature, channel concatenation is an effective way to reserve the two forms of information to the maximum extent.

C. Probability-Guided Body Part Partition

During walking, different human parts usually reveal distinct moving patterns. For instance, the human torso remains relatively stable while the legs or arms swing rapidly. Besides, every person has his unique walking habit and swings each body part in his distinct way. Hence, many vision-based methods focus on extracting the unique part features of each body region, which is beneficial for precise gait recognition. Typically, most of them usually split the human body feature into different body parts [37], [38]. The human gait images are first fed into deep CNNs to extract high-level feature maps. For visual images, the height of the image is proportional to the spatial height of the human. Thus, they split the feature map along the height axis uniformly to get each body part's feature.

However, the radar point cloud is sparse in contrast to the visual image. There are significant differences among human body parts in terms of RCS, and body parts with higher RCS will produce more points. For this reason, the human torso will occupy a larger feature area than the legs or arms. As shown in Fig. 6, splitting the point feature uniformly is not equivalent to an equal partition of human height dimension. To divide the human body equally with radar point clouds, we propose a probability-guided body part partition strategy. First, we analyze the height distribution of the radar point cloud. Specifically, the point number within each spatial region is counted based on our training data. From Fig. 6, we can discover that the number of points induced by the human torso is the largest while the head and feet produce the least points. Similar to vision-based methods, our method divides the spatial region

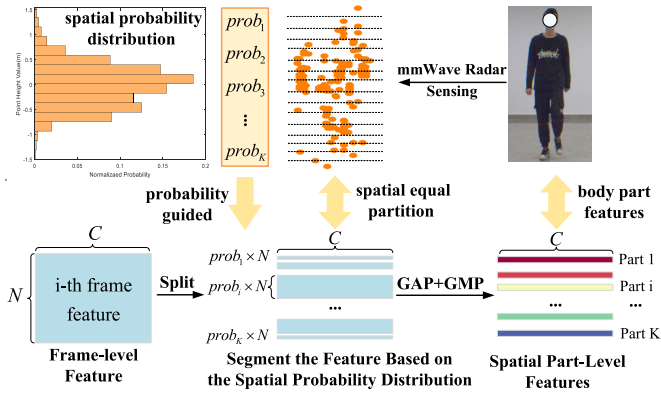


Fig. 6. Detailed structure of body partition module. Based on the spatial distribution probability of the radar point along the height axis, we split the frame-level features unequally to produce the feature of each body part. N indicates the number of points, C indicates the number of feature channels, and K indicates the number of body parts.

equally along the height dimension. The probability of each part is obtained

$$\text{prob} = \{\text{prob}_1, \text{prob}_2, \dots, \text{prob}_K\} \quad (7)$$

where $\text{prob}_k \in [0, 1]$ indicates the probability of k th part, K denotes the number of parts, and $\sum_{k=1}^K \text{prob}_k = 1$. As mentioned above, points within each frame P_i have been sorted by their height value, while the subsequent pointwise feature extraction module does not affect the order of points. Consequently, the row dimension of frame-level feature $F_i \in \mathbb{R}^{N \times C}$ is also relevant to the spatial height of the human body. Based on the statistical probability, F_i is split into K different regions along row dimension

$$\{F_i^{\text{region}1}, F_i^{\text{region}2}, \dots, F_i^{\text{region}K}\} \quad (8)$$

where $F_i^{\text{region}k} \in \mathbb{R}^{N_k \times C}$, $k = 1, 2, \dots, K$ denotes the feature of k th region, $N_k = N \times \text{prob}_k$ denotes the height of $F_i^{\text{region}k}$. In this way, the frame-level feature is split according to the spatial distribution of human body parts. An equal partition strategy may divide different body parts into the same region, leading to a coarse representation of gait features. In contrast, our partition strategy can split the feature finely and obtain more fine-grained spatial gait features.

Moreover, to aggregate the feature within each region, global max pooling and average pooling are applied

$$F_i^k = \text{GMP}(F_i^{\text{region}k}) + \text{GAP}(F_i^{\text{region}k}) \quad (9)$$

where $F_i^k \in \mathbb{R}^C$ denotes the feature vector of the k th part.

D. Local-Global Temporal Feature Extraction

For each part feature vector F_i^k , by gathering feature vector over T frames, we can obtain the time-varying part feature $B_k = \{F_i^k | i = 1, \dots, T\} \in \mathbb{R}^{T \times C}$. Generally, gait pattern changes dynamically, and a single frame could only reflect limited gait information. We are supposed to model temporal variation among consecutive frames for discriminative gait features. Currently, many vision-based methods apply temporal convolution along the time dimension to extract temporal clues

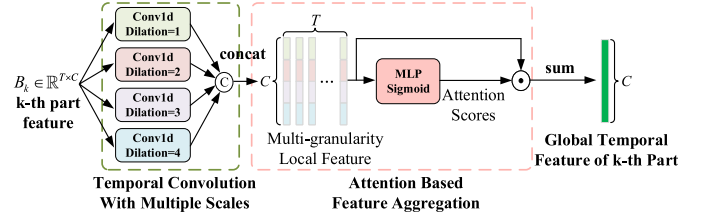


Fig. 7. Detailed structure of proposed LGTE module. Our LGTE consists of temporal convolution with multiple scales and attention-based feature aggregation. T indicates the number of frames and C indicates the number of feature channels.

from gait sequence [39]. Compared with LSTM, temporal convolution is lightweight and easy to optimize. Besides, temporal convolution is suitable for extracting local temporal features within a short term. Inspired by that, we apply multi-scale temporal convolution to model the local temporal clues among frame sequences. In contrast to single convolution, our multiscale convolution enriches the diversity of local temporal features, leading to more fine-grained temporal gait representation.

As shown in Fig. 7, for input k th part feature $B_k \in \mathbb{R}^{T \times C}$, we design a four branches temporal convolution architecture with different receptive fields

$$\begin{aligned} B_k^1 &= \text{Conv1}(B_k), & B_k^2 &= \text{Conv2}(B_k) \\ B_k^3 &= \text{Conv3}(B_k), & B_k^4 &= \text{Conv4}(B_k) \end{aligned} \quad (10)$$

where $B_k^1, B_k^2, B_k^3, B_k^4 \in \mathbb{R}^{T \times (C/4)}$ indicates the feature of each temporal granularity, respectively. To preserve the temporal information of each granularity, we cascade four features along channel dimension and get $B_k^{\text{cat}} = [B_k^1 || B_k^2 || B_k^3 || B_k^4] \in \mathbb{R}^{T \times C}$, where $[||]$ is the channel concatenation operation. Concretely, we apply a dilation convolution kernel rather than a normal convolution kernel. Dilation convolution is able to realize a larger temporal receptive field with fewer parameters and computation cost, which can alleviate the overfitting phenomenon.

Moreover, it's worth noting that the variation of gait patterns within different temporal granularities is totally distinct. For instance, the swing of legs within a short-term duration represents the tiny motion characteristic while the swing within a long-term duration represents the periodic characteristic. To aggregate the temporal features of different granularities and produce the global temporal feature of each body part, we propose an attention-based summation strategy. Specifically, we design an attention module to calculate the importance weight of each frame

$$B_k^{\text{atten}} = \text{Sigmoid}(\text{MLP}(B_k^{\text{cat}})) \quad (11)$$

where $B_k^{\text{atten}} \in \mathbb{R}^{T \times C}$ denotes the importance score, $B_{ki}^{\text{atten}} \in \mathbb{R}^C$ denotes the score of i th frame. Our attention score B_{ki}^{atten} is a feature vector rather than a scalar, which is able to fuse multigranularity local temporal features adaptively. After that, we multiply every frame with its attention dimension and sum the weighted feature along the temporal dimension

$$B'_k = \frac{\sum_{i=1}^T B_{ki}^{\text{atten}} \odot B_{ki}^{\text{cat}}}{\sum_{i=1}^T B_{ki}^{\text{atten}}} \quad (12)$$

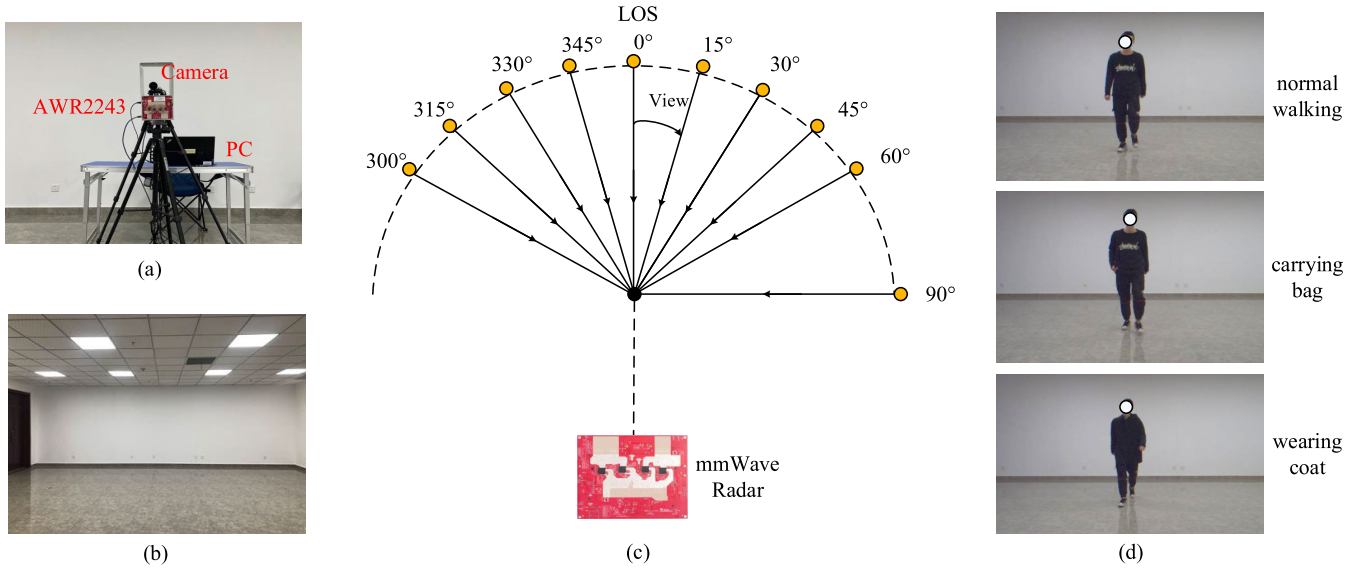


Fig. 8. Diagram of data collection (a) platform, (b) scene, (c) ten walking views, and (d) three wearing conditions.

where $B'_k \in \mathbb{R}^C$ indicates the output global part-level feature of k th part, $\odot$ is the dot product operation. In comparison with max pooling, which could only preserve the maximum value among frames, our weighted summation method can highlight or suppress temporal features adaptively. Gathering the K part-level features, the final output feature $B' = \{B'_k | k = 1, \dots, K\} \in \mathbb{R}^{K \times C}$ is obtained. In consideration of the great difference between body parts, the convolution parameter of each part is independent. In other words, the feature extraction manner of each body part is distinct, which is beneficial for fine-grained temporal feature extraction.

E. Loss Function

For the retrieval task, we employ batch all triplet loss [40] to optimize our network. Triplet loss aims to minimize the intrainity discrepancy and maximize the interidentity discrepancy. The output feature of samples within a batch is used to calculate the triplet loss

$$\mathcal{L}_{BA}(\theta; X) = \sum_{i=1}^P \sum_{a=1}^Q \sum_{\substack{p=1 \\ p \neq a}}^Q \sum_{\substack{j=1 \\ j \neq i}}^Q \sum_{n=1}^Q [\max(m + D_p - D_n, 0)] \quad (13)$$

where P denotes the number of sampled identities in a batch and Q denotes the number of sampled sequences for each identity. $D(\cdot)$ denotes the Euclidean distance, $D_p = D(B'_a, B'_p)$ is the distance of features from the same identity, $D_n = D(B'_a, B'_n)$ is the distance of features from different identities, and m denotes the margin threshold which controls the distance between D_p and D_n .

V. EXPERIMENTS AND ANALYSIS

A. Experiment Setup and Data Collection

As shown in Fig. 8(a), we have set up a data collection platform with a mmWave radar (AWR2243), a camera, and

TABLE I
RADAR PARAMETERS

Parameter	Value
Carrier Frequency	77 GHz
Chirp Duration	15.5 us
Chirp Slope	100 MHz/us
Bandwidth	1200 MHz
Samples Per Chirp	128
Sampling Frequency	12.5 Msps
Chirps Per Frame	128
Frame Duration	25 ms

a PC. The camera is used for scene record. For completely irradiating the human body, radar is placed 1.1 m above the ground, which is close to the middle height of the human torso. And we collect human walking data in an empty room shown in Fig. 8(b).

The mmWave radar AWR2243 is equipped with a MIMO antenna array which consists of 12 transmitters and 16 receivers. Besides, it works in time-division multiplexing multiple-input-multiple-output (TDM-MIMO) mode, which means the 12 transmitters emit frequency-modulated sinusoid waves successively. As a result, our radar data cube contains 192 antenna channels. The azimuth angular resolution is 1.4° and the elevation angular resolution is 18° . The high angular resolution enables more accurate direction estimation. The details of radar parameters employed in our experiment are listed in Table I.

Since there's a conflict between each index, we design the parameters rigorously in consideration of human walking velocity and range. The final range resolution is 0.125 m, and the maximum unambiguous range is 18.75 m. Our maximum range meets the requirement of recognition at a certain distance. The velocity resolution is 0.082 m/s, maximum unambiguous velocity is 5.24 m/s. According to our observation, the maximum velocity of human feet may sometimes

reach 5 m/s, and our system can measure the velocity of every body part unambiguously.

For vision-based gait recognition research, there are many public datasets such as CASIA-B [41] and OU-MVLP [42], both of which contain gait data under different walking views and wearing conditions. However, currently, there is no public radar point cloud dataset containing multiwalking views or multiwearing conditions data. To evaluate the performance of radar point cloud-based gait recognition model under realistic scenarios, following the data collection settings of vision dataset, as shown in Fig. 8(c) and (d), we have collected radar data from 90 subjects under ten walking views (0° , 15° , 30° , 45° , 60° , 90° , 300° , 315° , 330° , and 345°) and three wearing conditions (normal walking, carrying bag, and wearing coat). Specifically, for each subject, we have collected ten groups of data under each walking view, which contain six groups of normal walking data (NM01–NM06), two groups of carrying bag data (BG01 and BG02), and two groups of wearing coat data (CT01 and CT02). On the whole, we have collected 100 groups of gait data for each subject and our dataset contains 9000 groups of gait data.¹ Our volunteers involve 47 males and 43 females, aged from 17 to 25 years, weight from 44 to 93 kg and measuring from 156 to 192 cm in height.

After collecting raw radar echo data, we follow the signal processing pipeline proposed in Section III to generate radar point cloud data for each subject. For dataset partition, we randomly select 55 subjects for training and 35 subjects for testing. And we do not further divide out the validation set. In the training stage, all samples are used to optimize the model. In the testing stage, in order to evaluate the model performance under cross-wearing conditions, we divide the first four groups of normal walking data (NM01–NM04) into gallery samples, and the remaining data (NM05 and NM06, BG01 and BG02, and CT01 and CT02) into probe samples.

B. Implementation Details

For each sample, we choose $T = 100$ number of frames as network input, which contains 2.5 s of walking data. The point number within each frame is fixed to $N = 128$ by randomly up-sampling or down-sampling. For up-sampling, we randomly select and copy points as paddings. For down-sampling, we randomly select and drop points. In order to eliminate the influence of absolute distance during network learning, the point 3-D coordinates are normalized by subtracting the centroid position of the point cloud within each frame. Besides, since the magnitude of point RCS is generally bigger than that of coordinates and velocity, the RCS value of every point is divided by the maximum RCS value within each frame. This normalization operation eliminates the magnitude difference among attributes.

In the dual-stream feature extraction module, for each pathway, we use a three-layer MLP to extract the point feature, of which the output feature channel is 32, 64, and 128 separately. And the output channel C of feature fusion MLP is set to 512. In the body part partition module, the output body parts K of max pooling and mean pooling is set as 16, and

the output channel is 512. In the temporal feature extraction module, we set the kernel size of temporal convolution as 3, while the dilation ratio of each temporal scale is set to 1, 2, 3, and 4, respectively. The output channel of each scale convolution is set to 128. In the temporal attention module, the kernel size of convolution is 3, and the output channel is 512. The output channel of the last FC is set as 256. The activation function of the network is LeakyReLU, of which the negative slope is set to 0.1.

In our training process, the AdamW optimizer is applied with a learning rate of 0.0001. Coefficients used for computing running averages of gradient and its square in AdamW are set to 0.9 and 0.999, respectively. Besides, L2 weight decay is applied for regularization, and the coefficient is 0.0001. The batch size is 8 with $P = 2$ and $Q = 4$. Triplet margin m is set to be 0.2. We train our network for 100k iterations with NVIDIA GTX 3080 GPU.

C. Evaluation

As mentioned above, our gait recognition system is based on a retrieval task instead of a classification task. We first calculate the similarity of the output feature between the probe sample and all gallery samples and then assign the identity of the most similar gallery sample to the probe sample. We report the rank-1 accuracy. Furthermore, we aim to investigate the model's performance under realistic scenarios. Hence we have conducted two types of experiment: cross-view and cross-wearing-condition. In other words, the walking view or wearing condition of the probe sample is not contained in the gallery set.

1) *Cross-View*: To evaluate the model's robustness to variation in walking view, we calculate the rank-1 accuracy for every walking view, respectively. For instance, when evaluating walking view 0° , we first select the probe samples with view 0° and exclude all gallery samples with view 0° . Then, we select the gallery samples with a specific view, e.g., 15° , and assign the identity to every probe sample by similarity measurement. More specifically, the view of the gallery sample is selected one by one, and the accuracy of each gallery view is calculated. The final recognition accuracy of view 0° is obtained by averaging the accuracy of the other nine gallery views.

2) *Cross-Wearing-Condition*: As mentioned above, our gallery set only contains samples with normal walking conditions, while our probe set contains all three wearing conditions. To evaluate the model's robustness to variation in wearing condition, we also calculate the rank-1 accuracy for every wearing condition, respectively. For instance, when evaluating wearing condition BG, we calculate the similarity between probe samples with bag carrying and all gallery samples. To be specific, we report the cross-view accuracy for every wear condition.

D. Comparison Results

1) *Compared With Radar-Based Networks*: To verify the effectiveness of our proposed network, we reproduce several existing radar point cloud-based gait recognition methods for

¹<https://github.com/LanDu-XD/mmPcGait>

TABLE II
RANK-1 ACCURACIES UNDER CROSS-VIEW AND CROSS-WEARING-CONDITION, EXCLUDING IDENTICAL-VIEW CASES

Gallery NM#1-4		Gallery (10 views)										Mean
Probe		0	15	30	45	60	90	300	315	330	345	
NM#5-6	SRPNet	28.25	37.36	35.90	36.83	27.46	22.22	28.89	35.87	34.49	32.37	31.96
	STPGCN	36.41	36.32	40.36	36.31	25.28	6.26	21.14	32.42	33.92	34.26	30.27
	HDNet	36.94	39.60	43.82	35.70	33.59	11.60	25.60	35.28	38.10	38.21	33.84
	PcGaitSet	49.84	50.72	55.38	51.59	51.43	44.76	46.19	53.02	50.92	54.11	50.80
	PcGaitPart	82.34	86.25	87.08	87.81	75.93	51.59	72.77	87.65	85.65	86.18	80.39
	PcCSTL	86.48	91.16	89.46	90.59	84.08	61.39	81.26	89.86	91.21	90.40	85.62
	ours	88.37	92.36	92.20	93.20	87.04	68.28	83.88	92.62	91.35	92.05	88.14
BG#1-2	SRPNet	30.00	28.03	27.70	30.95	29.52	15.78	17.62	27.14	29.52	31.11	26.74
	STPGCN	38.82	33.09	30.79	28.17	24.10	6.78	16.17	27.39	30.57	33.30	26.92
	HDNet	35.71	29.85	32.60	28.43	27.14	11.99	21.98	22.78	31.25	30.20	27.19
	PcGaitSet	48.10	36.65	38.81	42.38	37.46	30.43	38.41	44.60	44.78	42.38	40.40
	PcGaitPart	81.54	74.48	77.03	77.58	67.36	36.76	55.35	71.10	76.41	76.85	69.45
	PcCSTL	88.34	80.94	80.12	84.05	76.18	47.06	68.12	77.07	82.90	84.07	76.89
	ours	90.43	84.37	85.17	88.56	79.88	61.38	69.26	81.90	84.20	85.53	81.07
CT#1-2	SRPNet	27.62	31.89	29.68	32.86	27.46	19.53	30.00	29.05	31.56	29.95	28.96
	STPGCN	37.91	31.40	30.39	31.86	24.44	7.88	20.62	30.40	30.16	28.96	27.40
	HDNet	32.06	32.63	38.02	32.61	24.24	16.76	23.54	25.43	32.78	32.61	29.07
	PcGaitSet	45.24	47.83	44.44	40.16	46.03	32.15	38.10	48.10	48.63	45.89	43.66
	PcGaitPart	81.54	79.36	79.72	82.89	71.24	42.03	67.81	75.77	76.52	77.00	73.39
	PcCSTL	85.57	85.05	85.17	85.40	80.81	55.83	78.73	82.00	85.27	85.16	80.90
	ours	88.69	85.51	86.21	86.55	81.10	62.67	75.93	83.91	86.37	86.17	82.31

comparison, namely, “SRPNet” [21], “STPGCN” [18], and “HDNet” [33]. Expect for [21], other methods are designed for classification tasks. Hence, we modify the output channel of the last FC layer within these classification-based methods and replace the original cross-entropy loss with triplet loss for the retrieval task. As shown in Table II, our method achieves obvious improvement over current radar-based methods under all cases, which demonstrates the robustness and superiority of the proposed methods. As mentioned above, current methods often employ complex point feature extraction and temporal feature extraction modules, which may be sufficient for ideal classification tasks. However, for realistic retrieval tasks under cross-view and cross-wearing conditions, their methods are vulnerable and redundant. Specifically, STPointGCN and HDNet apply point-based graph convolution to extract point-level features, which aims to explore the local geometric structure information. Nevertheless, geometric relation among points varies significantly when the walking view changes, making it a suboptimal option for robust feature extraction. Besides, SRPNet and HDNet apply LSTM and Transformer, respectively, to extract temporal feature, which lacks the capacity to model motion patterns within a short time scale. Furthermore, they treat all body points as a whole and neglect the uniqueness of each body part, leading to coarse-grained gait representation.

In terms of the walking view, the accuracy of 90° is the worst among the ten walking views. That’s because the Doppler velocity measured by radar will degrade seriously

when the walking direction of the subject is perpendicular to radar LOS, and the appearance information varies greatly from 90° to other walking views. Meanwhile, middle views such as 45° and 30° achieve better performance in contrast to 0°. The reason is that the difference between middle views and other views is less than 0°. Besides, since our gallery set only contains samples under normal walking, the accuracy usually drops when the wearing condition of the probe is different from gallery samples. The accuracy under normal walking conditions is usually the best.

2) *Compared With Visual-Based Networks:* As discussed above, current radar-based methods have several drawbacks. To further evaluate the effectiveness of the proposed methods, we reproduce some camera-based gait recognition methods for comparison, namely, “GaitSet” [37], “GaitPart” [38], and “CSTL” [39]. Since these methods typically employ vanilla CNN to extract features from visual images. For a fair comparison, we replace the front CNN within these methods with our MLP to extract features from the radar point cloud, while the rest of their architecture remains unchanged. Specifically, these methods split the body features uniformly and design their own module to extract temporal features for each body part. We prefix these modified architectures with “Pc.” As shown in Table II, our proposed method still outperforms these visual-based networks under cross-view and cross-wearing conditions. Since these methods are originally designed for retrieval tasks they take into account the variation in walking view and wearing condition. They

TABLE III
ABLATION STUDY

	DSFE	PGBP	LGTE	NM	BG	CT	Mean
Baseline	-	-	-	67.31	54.29	59.94	60.51
Ours	✓			70.17	59.86	63.47	64.50
		✓		79.84	70.30	72.62	74.25
			✓	78.35	70.47	72.42	73.75
	✓	✓		81.36	70.58	74.08	75.34
	✓		✓	78.37	69.89	72.25	73.50
		✓	✓	87.46	78.92	81.19	82.52
	✓	✓	✓	88.14	81.07	82.31	83.84

present better performance than existing radar-based methods to some extent. However, the accuracy of these methods is still unsatisfactory in two aspects. First, their uniform partition strategy is not suitable for sparse radar point clouds. Second, the diversity of their temporal feature is defective. GaitSet proposes Set Pooling to aggregate temporal information, which only calculates the average, median, and maximum value within different frames. Although GaitPart has considered the short-range spatial-temporal representations, their time scale is single and limited. CSTL proposes to extract frame-level, short-term and long-term features, and thus achieve better performance. By contrast, our method splits the body feature according to the property of the radar point cloud. Besides, we extract temporal features in multiple scales and aggregate them adaptively. Consequently, our method achieves 2.52%, 4.18%, and 1.41% improvement over PcCSTL under three wearing conditions. Since one arm of the subject keeps nearly stationary under carrying bag conditions, the gait information is much less than in other conditions. Notably, our proposed method gains great advancement under carrying bag conditions (from 76.89% to 81.07%), which further demonstrates the effectiveness of the proposed methods.

E. Model Analysis

1) *Ablation Study*: To further investigate the effectiveness of each module in our network, extensive ablation experiments are conducted. The results are shown in Table III, where DSFE, PGBP, and LGTE present the DSFE module, PGBP module, and LGTE module, respectively. The baseline model contains none of the three modules. That is, we apply a single-stream architecture to extract pointwise features and aggregate all pointwise features by max pooling, which means all points are handled as a whole. Besides, the baseline model applies a single-scale temporal convolution to extract temporal features and aggregate all frame features by max pooling. Several conclusions are summarized as follows.

- 1) Compared with the baseline model, all three modules can improve the model accuracy. The combination of the three modules achieves the best performance (from 60.51% to 83.84% in terms of average accuracy), which verifies the effectiveness and complementarity of the proposed modules.

TABLE IV
STUDY OF THE EFFECTIVENESS OF DUAL-STREAM ARCHITECTURE

Architecture	NM	BG	CT	Mean
Single-stream with Velocity	59.78	52.44	53.68	55.30
Single-stream with XYZ+RCS	80.28	69.31	72.64	74.08
Single-stream with All Attributes	87.46	78.92	81.19	82.52
Dual-stream	88.14	81.07	82.31	83.84

- 2) Compared with the baseline model, applying DSFE offers a 3.99% improvement. Besides, when integrating with other modules, DSFE still gains performance enhancement, which demonstrates the generality of the proposed point feature extraction strategy.
- 3) PGBP and LGTE are vital components for fine-grained feature extraction, which gains 13.74% and 13.24% improvement, respectively, over the baseline model in terms of mean accuracy. PGBP enables the model to learn fine-grained spatial features from the human body point cloud. Average accuracy is low (73.50%) without body partition, demonstrating the uniqueness of each body part. Besides, single-scale temporal convolution is insufficient for fine-grained feature extraction, LGTE enables the model to learn fine-grained temporal features from various frames. Applying both PGBP and LGTE enables the model to extract fine-grained spatial-temporal features.

2) *Analysis of Point Feature Extraction Strategies*:

In order to analyze the impact of point feature extraction strategies, we first explore the effectiveness of our proposed dual-stream architecture. The results are given in Table IV.

- 1) The performance of the spatial-velocity pathway (single-stream with velocity) is inferior to that of the spatial-RCS pathway (single-stream with XYZ + RCS), which indicates the information of 3-D coordinates and RCS holds greater significance for gait recognition than velocity alone.
- 2) With only limited information as input (velocity or XYZ + RCS), the recognition accuracy drops a lot.

TABLE V
STUDY OF THE EFFECTIVENESS OF MLP

Framework	NM	BG	CT	Mean
RSCNN	65.07	57.29	58.77	60.38
DGCNN	77.03	66.12	70.38	71.18
AdaptNet	82.43	71.54	75.30	76.62
DSFE	88.14	81.07	82.31	83.84

However, by incorporating all attributes of the point, the accuracy achieves great improvement, demonstrating the complementarity of velocity and 3-D coordinates.

- 3) Single-stream with all attributes as input treats five attributes of the point as a whole, which is coarse-grained and insufficient. Compared with single-stream, dual-stream architecture can effectively enhance the performance of point feature extraction, which demonstrates the difference among point attributes and the necessity of two pathways.

Similar to PointNet, we apply MLP instead of a complex point feature extraction module to independently extract point-wise features, which is simple but effective. To investigate the effectiveness of MLP, we replace MLP with other popular point feature extraction modules and evaluate the recognition performance. The results are shown in Table V, where DGCNN [43] and AdaptNet [44] apply graph convolution operation for point feature extraction, RSCNN [45] applies relation-shape point convolution operation. Unlike experiment results in the visual point cloud classification task, the performance of this complex architecture is unsatisfactory in our gait recognition task. For instance, RSCNN overly relies on local geometric relations and thus performs poorly for sparse radar point clouds. DGCNN and AdaptNet treat every point as the node of a graph and mainly rely on the geometric topotaxy between points. Although they perform better than RSCNN, their results are still unsatisfactory. By contrast, our proposed dual-stream MLP framework can effectively extract features from the radar point cloud.

3) *Analysis of Body Partition Strategies*: At present, there is no body part partition strategy designed for sparse radar point clouds. To explore the impact of the body partition strategy, we compare our probability-guided body partition with uniform partition and none partition. The results are given in Table VI. It's clearly discovered that body partition is a crucial step for gait recognition. None partition only leads to an average accuracy of 73.50%, while partition strategy significantly improves the accuracy. Besides, our PGBP outperforms other partition strategies to a great extent, particularly under carrying bag conditions (from 77.74% to 81.07%), which demonstrates the suitability of the proposed partition strategy for sparse radar point clouds. In contrast, the uniform partition may divide body parts improperly because of the sparsity of the radar point cloud. Points within different spatial

TABLE VI
STUDY OF THE EFFECTIVENESS OF BODY PART PARTITION STRATEGIES

Partition Strategy	NM	BG	CT	Mean
None Partition	78.37	69.89	72.25	73.50
Uniform Partition	87.23	77.74	81.00	81.99
PGBP	88.14	81.07	82.31	83.84

TABLE VII
IMPACT OF THE NUMBER OF PARTITION PARTS

Parts	NM	BG	CT	Mean
4	83.20	77.69	78.36	79.75
8	87.77	80.53	82.22	83.50
16	88.14	81.07	82.31	83.84
24	86.16	77.51	79.54	81.07

areas may be divided into the same region, leading to limited performance.

Besides, the number of partition parts will also affect recognition accuracy. To further investigate the impact of it, we attempt to divide body features into different numbers of parts, and the results are given in Table VII. Compared with none partition, the accuracy under all three conditions increases as the partition parts rise. Concretely, dividing the human body into 16 parts promotes the average accuracy by 3.75% in contrast to four parts, which proves the significance of fine-grained spatial part modeling. Viewing the whole body as four parts is a coarse-grained partition strategy. Besides, it's worth noting that the growth from eight parts to 16 parts is 0.34%, and the accuracy drops by 2.77% from 16 parts to 24 parts. This phenomenon indicates that the 16 parts is an appropriate partition number and dividing too many parts may not achieve large improvement.

4) *Analysis of Temporal Feature Extraction Strategies*:

In our experiment settings, we apply four scales of temporal convolution to extract fine-grained temporal gait features. To further explore the impact of the number of temporal scales, we, respectively, apply different scales of temporal convolution and analyze the recognition performance, the results are shown in Table VIII. It can be noticed that multiscale temporal features achieve better performance compared with single-scale features. Among them, single-scale temporal features could only represent limited gait characteristics, leading to poor average accuracy. As the temporal scales rise, the average accuracy increases significantly (from 79.65% to 83.84%), which indicates the effectiveness of multiscale temporal modeling. Besides, the average accuracy drops by 0.65% from four scales to five scales, this phenomenon indicates that more temporal scales may bring more model parameters and may not achieve better performance. For our gait recognition task,

TABLE VIII
IMPACT OF THE NUMBER OF TEMPORAL SCALES

Scales	NM	BG	CT	Mean
1	84.05	75.93	78.65	79.65
2	86.22	77.77	80.91	81.63
3	86.56	78.16	80.96	81.89
4	88.14	81.07	82.31	83.84
5	87.42	80.02	82.13	83.19

TABLE IX
STUDY OF THE EFFECTIVENESS OF TEMPORAL
FEATURE EXTRACTION METHODS

Methods	NM	BG	CT	Mean
LSTM	73.97	66.74	66.62	69.11
Transformer	81.06	71.41	75.46	75.98
LGTE	88.14	81.07	82.31	83.84

TABLE X
STUDY OF THE EFFECTIVENESS OF TEMPORAL
FEATURE AGGREGATION STRATEGIES

Methods	NM	BG	CT	Mean
Mean Pooling	73.95	61.44	65.75	67.05
Max Pooling	84.12	74.88	77.48	78.82
Ours	88.14	81.07	82.31	83.84

four scales of temporal granularities are sufficient for feature representation.

LSTM and transformer are two popular temporal feature extraction methods. To investigate the impact of the temporal feature extraction module, we replace our LGTE module with LSTM and Transformer, the results are given in Table IX. As seen, our proposed LGTE module outperforms commonly used LSTM and transformer by a large margin (14.43% and 7.86% in terms of average accuracy). LSTM and transformer architecture mainly focus on long dependency within time series and lack the ability to extract temporal features in short temporal scales. In contrast, our proposed LGTE is helpful in capturing fine-grained gait patterns in diverse temporal scales. Moreover, LSTM and transformer architecture are generally hard to optimize compared with 1-D temporal convolution.

In order to investigate the effects of the frames aggregating strategy, we compare our method with max pooling and average pooling. The results are shown in Table X. The comparison between mean pooling and max pooling declares that simply aggregating time sequences by summation is not sufficient. The importance of each frame is different, as an easy strategy, max pooling can select the feature with maximum value and ignore the feature with low value. However, simply choosing the maximum feature may induce information loss, in contrast, our proposal can model the importance of

TABLE XI
SPACE AND TIME COMPLEXITY OF EACH PROPOSED MODULE

Model	#Params(M)	FLOPs(G)	Inference time(ms)
w.o. DSFE	8.08	3.07	13.55
w.o. PGBP	8.12	3.76	13.62
w.o. LGTE	4.45	3.40	8.71
Ours	8.13	3.77	14.69

each frame and aggregate them by weighted summation. Our aggregation strategy achieves 5.02% accuracy enhancement than max pooling, which further proves the superiority of our attention-based aggregation.

5) *Analysis of Model Complexity*: To explore the complexity of each proposed fine-grained feature extraction module, we, respectively, remove each module from the network and analyze the complexity, the results are shown in Table XI. Specifically, the parameters of the recognition model are counted to evaluate the space complexity, and the floating point operations (FLOPs) and inference time (ms) of the recognition model are counted to evaluate the time complexity. All experiments are conducted on a single GTX 3080 GPU, and we calculate the inference time by averaging the inference time of all test samples.

- 1) To investigate the impact of DSFE, we apply single-stream architecture without any feature fusion to extract pointwise features to investigate the impact of DSFE. As our feature fusion strategy in DSFE is the simplest channel concatenation, dual-stream architecture increases the parameters(+0.5 M), FLOPs(+0.7G), and inference time(+0.14 ms) a little in contrast to single-stream.
- 2) To investigate the impact of PGBP, we replace our partition strategy with a uniform partition. Our PGBP only changes the partition pattern and hardly introduces new parameters(+0.1 M). Besides, PGBP merely increases the FLOPs(+0.01G) and inference time(1.07 ms) by a little bit.
- 3) To investigate the impact of LGTE, we replace multiscale temporal convolution with a single scale. For temporal feature aggregation, we simply apply max pooling rather than attention-based summation. Single-scale temporal convolution definitely reduces the model parameters compared with multiscale (+3.68 M). With an attention-based aggregation module, LGTE increases the FLOPs and inference time by 0.37G and 5.98 ms, respectively.

It's worth noting that our attention module applies convolution operation rather than self-attention to calculate attention scores. Compared with transformer encoder architecture, our attention module is lightweight and brings out complexity increments in an acceptable range. In conclusion, our proposed fine-grained feature extraction modules improve the performance of the recognition model by a large margin with an acceptable increment in model complexity.

TABLE XII
SPACE AND TIME COMPLEXITY OF RECOGNITION MODEL

Model	#Params(M)	FLOPs(G)	Inference time(ms)
SRPNet	7.39	4.50	177.81
STPGCN	1.08	2.43	160.52
HDNet	4.47	6.50	26.62
PcGaitSet	1.23	6.92	11.26
PcGaitPart	10.71	3.89	13.10
PcCSTL	30.66	5.89	13.63
Ours	8.13	3.77	14.69

Table XII indicates the complexity of our model and other comparison models.

- 1) The params of PcCSTL are obviously larger than other models. PcCSTL involves a complex temporal feature extraction module, which will inevitably bring out numerous parameters.
- 2) As the temporal feature extraction module of SRPNet and STPGCN is in sequential order, their inference time is longer than other models. To be specific, their temporal module depends on the previous state to generate the current state, which means we need to calculate the output of each time step in the sequence.
- 3) HDNet employs a frame sampling strategy and Transformer encoder to extract the gait feature, which also costs a certain FLOPs and inference time.
- 4) Compared with PcGaitPart and PcCSTL, our proposed model has fewer parameters and FLOPs. However, in our LGTE module, we need to successively extract temporal features in multiple scales, which will unavoidably increase the inference time. Consequently, the inference time of our model is a bit longer than PcGaitPart(+1.59 ms) and PcCSTL(+1.06 ms). From an overall perspective, the inference time of the proposed model is still within the same order of magnitude as that of the better-performing comparison methods.

Models with larger parameters require more memory space, and those with longer inference times take more time to recognize targets, both of which constrain the deployment of recognition systems. Taking space complexity, time complexity, and recognition accuracy into consideration, the practicability of our proposed model is still better than other methods.

6) *Analysis of View Discrepancy*: The discrepancy of walking view between probe and gallery samples affects the recognition accuracy greatly. To explore the zero-boundary condition of the walking view, we evaluate the average accuracy under different view discrepancies. As shown in Fig. 9, the accuracy degrades severely as the view discrepancy increases. Specifically, the average accuracy of the three methods is 99.86%, 99.86%, and 99.28%, respectively, when

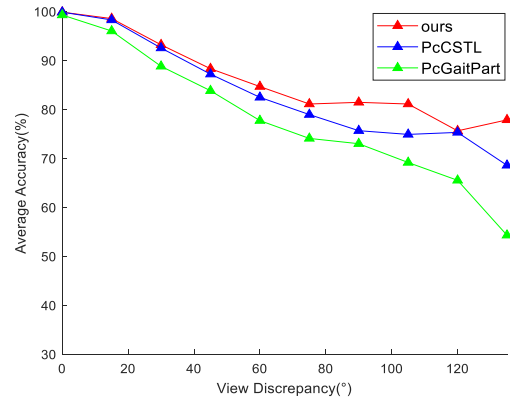


Fig. 9. Impact of the view discrepancy on average recognition accuracy.

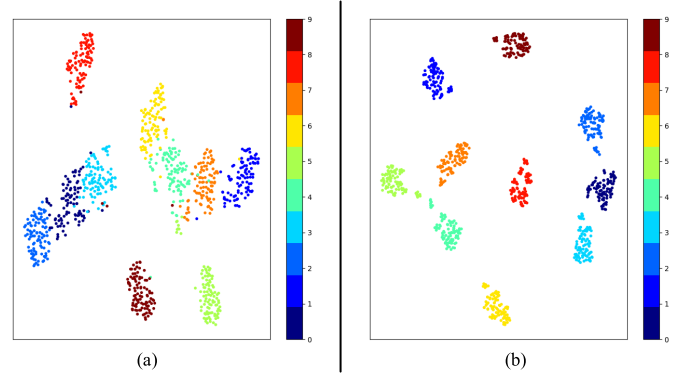


Fig. 10. Feature visualization of 10 identities via t-SNE. (a) Baseline model. (b) Our proposed model. Every identity is represented by a unique color and each point indicates a test sample.

the view of probe and gallery samples is the same. In contrast, the average accuracy is only 77.86%, 68.57%, and 54.29%, respectively, when the view discrepancy is 135°. Our method shows better robustness in viewing discrepancy compared with other methods. However, the accuracy of our method drops below 80% when the view difference is larger than 100°. In practical applications, to guarantee the model performance under large view discrepancy conditions, we are supposed to collect abundant views of samples to construct a gallery set.

7) *Visualization*: As mentioned above, our gait recognition model is based on a retrieval task and aims to minimize the intraindiscrepancy and maximize the interidentity discrepancy. In order to visually display the feature distributions of different identities, we randomly select ten identities from our dataset and apply t-SNE to visualize the output feature distribution. As shown in Fig. 10, the output feature of the baseline model is compact. The features of different identities gather in the same area, leading to poor identity separability. In contrast, the feature distributions of our proposed methods are more dispersive and thus are more distinguishable, which further proves the great feature representation ability of our method.

VI. CONCLUSION

In this article, we propose a novel gait recognition network to extract fine-grained spatial-temporal features from mmWave radar point cloud. For fine-grained spatial

feature extraction, we proposed a DSFE module to exploit the abundant information available in the radar point cloud. Moreover, we also propose a PGBP module to split sparse radar point clouds into fine-grained spatial body parts. For fine-grained temporal feature extraction, a multiscale temporal feature extraction (LGTE) module is proposed to capture the temporal patterns of each body part with multiple scales. To verify the effectiveness of our method, we have collected human walking data from 90 subjects under ten walking views and three wearing conditions. And we conduct extensive experiments in various realistic settings, i.e., cross-view and cross-wearing-condition. The results demonstrate the efficiency of the proposed three key modules, and the combination of them enables our network to extract more fine-grained and discriminative gait features from radar point clouds than existing methods.

In the future, we will continue to investigate radar-based gait recognition from two aspects. On the one hand, we only focus on single-person recognition in the current experiment. We plan to study gait recognition in multiperson situations, which involves robust multiperson tracking and recognition. On the other hand, to promote the practicability of our method, we plan to study gait recognition under more abundant gait types and environment settings. It would be an interesting and also more challenging issue to perform gait recognition in complex scenarios.

REFERENCES

- [1] S. Sarkar, P. J. Phillips, Z. Liu, I. R. Vega, P. Grother, and K. W. Bowyer, "The humanID gait challenge problem: Data sets, performance, and analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 2, pp. 162–177, Feb. 2005.
- [2] A. Sepas-Moghaddam, F. M. Pereira, and P. L. Correia, "Face recognition: A novel multi-level taxonomy based survey," *IET Biometrics*, vol. 9, no. 2, pp. 58–67, Mar. 2020.
- [3] K. Cao and A. K. Jain, "Automated latent fingerprint recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 788–800, Apr. 2019.
- [4] K. Nguyen, C. Fookes, R. Jillela, S. Sridharan, and A. Ross, "Long range iris recognition: A survey," *Pattern Recognit.*, vol. 72, pp. 123–143, Dec. 2017.
- [5] A. Sepas-Moghaddam and A. Etemad, "Deep gait recognition: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 264–284, Jan. 2023.
- [6] J. Mantyjarvi, M. Lindholm, E. Vildjiounaite, S. Makela, and H. Ailisto, "Identifying users of portable devices from gait pattern with accelerometers," in *Proc. ICASSP*, Mar. 2005, p. 973.
- [7] A. Kale et al., "Identification of humans using gait," *IEEE Trans. Image Process.*, vol. 13, no. 9, pp. 1163–1173, Sep. 2004.
- [8] J. Ahn, K. Nakashima, K. Yoshino, Y. Iwashita, and R. Kurazume, "2 V-gait: Gait recognition using 3D LiDAR robust to changes in walking direction and measurement distance," in *Proc. IEEE/SICE Int. Symp. Syst. Integr. (SII)*, Jan. 2022, pp. 602–607.
- [9] W. Wang, A. X. Liu, and M. Shahzad, "Gait recognition using WiFi signals," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, Sep. 2016, pp. 363–373.
- [10] B. Vandersmissen et al., "Indoor person identification using a low-power FMCW radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3941–3952, Jul. 2018.
- [11] X. Bai, Y. Hui, L. Wang, and F. Zhou, "Radar-based human gait recognition using dual-channel deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9767–9778, Dec. 2019.
- [12] J. Pegoraro, F. Meneghello, and M. Rossi, "Multiperson continuous tracking and identification from mm-wave micro-Doppler signatures," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 2994–3009, Apr. 2021.
- [13] M. A. Richards, *Fundamentals of Radar Signal Processing*. New York, NY, USA: McGraw-Hill, 2014.
- [14] A. Venon, Y. Dupuis, P. Vasseur, and P. Meriaux, "Millimeter wave FMCW RADARs for perception, recognition and localization in automotive applications: A survey," *IEEE Trans. Intell. Vehicles*, vol. 7, no. 3, pp. 533–555, Sep. 2022.
- [15] P. Zhao et al., "MID: Tracking and identifying people with millimeter wave radar," in *Proc. 15th Int. Conf. Distrib. Comput. Sensor Syst. (DCOSS)*, May 2019, pp. 33–40.
- [16] Z. Meng et al., "Gait recognition for co-existing multiple people using millimeter wave sensing," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 849–856.
- [17] T. Li, X. Cao, H. Liu, C. Shi, and P. Chen, "MTPGait: Multi-person gait recognition with spatio-temporal information via millimeter wave radar," in *Proc. IEEE 27th Int. Conf. Parallel Distrib. Syst. (ICPADS)*, Dec. 2021, pp. 660–666.
- [18] C. Wang, P. Gong, and L. Zhang, "StpointGCN: Spatial temporal graph convolutional network for multiple people recognition using millimeter-wave radar," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 3433–3437.
- [19] A. Graves, *Supervised Sequence Labelling With Recurrent Neural Networks*. Cham, Switzerland: Springer, 2012.
- [20] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–12.
- [21] Y. Cheng and Y. Liu, "Person reidentification based on automotive radar point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5101913, doi: [10.1109/TGRS.2021.3073664](https://doi.org/10.1109/TGRS.2021.3073664).
- [22] D. Lan, C. Xiaoyang, S. Yu, X. Shikun, and X. Meng, "MMRGait-1.0: A radar time-frequency spectrogram dataset for gait recognition under multi-view and multi-wearing conditions," *J. Radar*, vol. 12, pp. 1–14, Jan. 2023.
- [23] L. Yang, G. Li, M. Ritchie, F. Fioranelli, and H. Griffiths, "Gait classification based on micro-Doppler features," in *Proc. CIE Int. Conf. Radar (RADAR)*, Oct. 2016, pp. 1–4.
- [24] L. Yang, G. Chen, and G. Li, "Classification of personnel targets with baggage using dual-band radar," *Remote Sens.*, vol. 9, no. 6, p. 594, Jun. 2017.
- [25] Z. Chen, G. Li, F. Fioranelli, and H. Griffiths, "Personnel recognition and gait classification based on multistatic micro-Doppler signatures using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 669–673, May 2018.
- [26] Y. Kim and H. Ling, "Human activity classification based on micro-Doppler signatures using a support vector machine," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 5, pp. 1328–1337, May 2009.
- [27] M. Zenaldin and R. M. Narayanan, "Radar micro-Doppler based human activity classification for indoor and outdoor environments," *Proc. SPIE*, vol. 9829, pp. 364–373, Jan. 2016.
- [28] P. Cao, W. Xia, M. Ye, J. Zhang, and J. Zhou, "Radar-ID: Human identification based on radar micro-Doppler signatures using deep convolutional neural networks," *IET Radar, Sonar Navigat.*, vol. 12, no. 7, pp. 729–734, Jul. 2018.
- [29] Y. Yang, C. Hou, Y. Lang, G. Yue, Y. He, and W. Xiang, "Person identification using micro-Doppler signatures of human motions and UWB radar," *IEEE Microw. Wireless Compon. Lett.*, vol. 29, no. 5, pp. 366–368, May 2019.
- [30] H. Li, A. Mehul, J. Le Kernec, S. Z. Gurbuz, and F. Fioranelli, "Sequential human gait classification with distributed radar sensor fusion," *IEEE Sensors J.*, vol. 21, no. 6, pp. 7590–7603, Mar. 2021.
- [31] Y. Shi et al., "Robust gait recognition based on deep CNNs with camera and radar sensor fusion," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10817–10832, Nov. 2023.
- [32] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 77–85.
- [33] Y. Huang et al., "HDNet: Hierarchical dynamic network for gait recognition using millimeter-wave radar," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
- [34] S. M. Patole, M. Torlak, D. Wang, and M. Ali, "Automotive radars: A review of signal processing techniques," *IEEE Signal Process. Mag.*, vol. 34, no. 2, pp. 22–35, Mar. 2017.
- [35] G. Chechik, V. Sharma, U. Shalit, and S. Bengio, "Large scale online learning of image similarity through ranking," *J. Mach. Learn. Res.*, vol. 11, no. 3, pp. 1109–1135, 2010.

- [36] C. Feichtenhofer, H. Fan, J. Malik, and K. He, "SlowFast networks for video recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6201–6210.
- [37] H. Chao, Y. He, J. Zhang, and J. Feng, "Gaitset: Regarding gait as a set for cross-view gait recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, 2019, pp. 8126–8133.
- [38] C. Fan et al., "GaitPart: Temporal part-based model for gait recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14213–14221.
- [39] X. Huang et al., "Context-sensitive temporal feature learning for gait recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12889–12898.
- [40] A. Hermans, L. Beyer, and B. Leibe, "In defense of the triplet loss for person re-identification," 2017, *arXiv:1703.07737*.
- [41] S. Yu, D. Tan, and T. Tan, "A framework for evaluating the effect of view angle, clothing and carrying condition on gait recognition," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, 2006, pp. 441–444.
- [42] N. Takemura, Y. Makihara, D. Muramatsu, T. Echigo, and Y. Yagi, "Multi-view large population gait dataset and its performance evaluation for cross-view gait recognition," *IPSI Trans. Comput. Vis. Appl.*, vol. 10, no. 1, Dec. 2018.
- [43] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, Oct. 2019.
- [44] H. Zhou, Y. Feng, M. Fang, M. Wei, J. Qin, and T. Lu, "Adaptive graph convolution for point cloud analysis," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4945–4954.
- [45] Y. Liu, B. Fan, S. Xiang, and C. Pan, "Relation-shape convolutional neural network for point cloud analysis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8887–8896.



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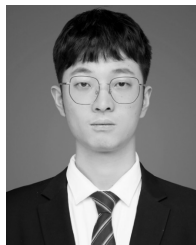
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